

LYUBASHIVKA, Ukraine

WMO: 337610

Lat: 47.8508N

Lon: 30.2683E

Elev: 600

StdP: 14.38

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	1.7	7.2	-3.9	4.6	3.2	1.0	5.9	8.5	23.3	20.7	21.3	22.9	6.4	0	0.496

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.8	90.5	68.3	87.3	67.0	84.4	66.1	71.1	84.3	69.5	82.4	68.1	80.4	6.9	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
67.0	101.4	76.2	65.3	95.7	74.4	63.9	90.8	73.1	35.3	84.6	33.9	82.6	32.7	80.3	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
18.9	16.9	14.4	-2.9	94.1	6.5	3.7	-7.5	96.8	-11.3	98.9	-14.9	101.0	-19.6	103.7		
			-3.4	75.2	6.4	4.2	-8.0	78.2	-11.8	80.7	-15.4	83.1	-20.1	86.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	49.3	25.4	28.8	37.6	50.3	60.8	67.7	71.3	70.4	60.5	48.9	39.4	29.5	
	DBStd	18.00	9.69	9.99	8.49	7.07	6.98	5.84	5.37	6.03	7.14	8.09	8.71	9.57	
	HDD50	2922	761	593	389	80	6	0	0	0	7	118	332	636	
	HDD65	6303	1226	1013	849	443	168	39	9	17	170	501	767	1101	
	CDD50	2680	0	0	5	88	340	531	661	634	321	85	15	0	
	CDD65	587	0	0	0	2	37	120	206	186	34	2	0	0	
	CDH74	5632	0	0	0	27	401	1061	1915	1872	342	14	0	0	
	CDH80	1881	0	0	0	5	88	298	683	728	79	0	0	0	
Wind	WSAvg	6.8	7.5	7.7	8.4	7.7	6.4	5.8	5.6	5.8	6.0	6.0	7.2	7.3	
Precipitation	PrecAvg	20.7	1.3	1.1	1.1	1.4	2.0	2.8	2.7	1.8	1.9	1.6	1.6	1.4	
	PrecMax	26.3	3.4	2.6	2.4	2.9	4.7	6.9	6.9	4.3	5.3	4.8	4.1	4.1	
	PrecMin	15.4	0.2	0.1	0.1	0.3	0.3	0.6	0.7	0.3	0.0	0.0	0.0	0.0	
	PrecStd	3.3	0.8	0.6	0.6	0.7	1.1	1.3	1.7	1.1	1.5	1.1	1.0	1.1	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	46.8	53.5	64.6	77.6	86.8	89.9	94.2	95.2	86.6	75.7	64.0	51.6	
		MCWB	42.2	45.9	50.3	59.1	63.5	67.7	69.0	69.5	64.3	60.6	55.2	48.5	
	2%	DB	42.7	47.2	58.6	71.3	82.3	86.4	90.3	90.9	81.9	70.5	57.8	48.1	
		MCWB	40.1	42.8	47.5	55.3	62.6	66.4	68.9	68.3	62.1	57.9	52.9	45.9	
	5%	DB	39.5	43.5	54.2	67.2	78.7	83.5	86.9	87.4	77.8	65.3	54.4	44.9	
		MCWB	37.8	40.6	44.8	53.0	61.3	66.1	67.8	67.1	60.9	56.0	51.5	43.3	
	10%	DB	37.0	40.5	50.2	63.6	74.7	80.5	83.7	83.9	73.8	60.9	51.4	41.6	
		MCWB	35.6	38.3	42.7	51.3	59.7	65.0	66.9	65.6	59.4	54.7	49.0	40.3	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	43.4	47.2	51.7	60.9	67.3	72.4	73.1	73.6	67.8	63.0	56.4	49.5
			MCDB	45.4	51.6	61.8	76.7	80.1	83.0	85.3	87.9	80.8	71.3	60.5	50.7
2%		WB	40.5	43.4	48.5	57.1	64.7	69.8	71.1	71.0	64.9	60.0	53.9	46.0	
		MCDB	42.4	46.4	56.6	68.1	77.0	81.1	84.1	85.7	76.2	66.9	56.9	47.3	
5%		WB	37.9	40.9	46.0	54.8	63.0	68.2	69.7	69.0	63.0	57.6	51.8	43.3	
		MCDB	39.2	43.0	52.9	63.9	75.3	79.2	82.8	82.4	73.0	63.7	54.1	44.8	
10%		WB	35.7	38.5	43.5	52.6	61.1	66.5	68.3	67.2	61.0	55.1	49.2	40.4	
		MCDB	36.8	40.5	48.7	61.2	72.0	76.8	80.7	79.8	70.8	60.4	51.2	41.6	
Mean Daily Temperature Range	5% DB	MDBR	8.3	9.9	14.1	18.6	19.6	19.1	19.8	20.6	18.1	15.0	9.9	8.5	
		MCDBR	9.2	14.1	21.3	24.7	24.7	23.0	24.2	25.1	24.6	21.5	13.8	10.5	
		MCWBR	7.5	9.9	12.3	12.2	10.2	9.0	8.4	9.3	10.5	11.2	8.9	8.5	
	5% WB	MCDBR	8.7	12.9	19.8	21.4	21.9	20.1	21.2	22.0	20.7	18.3	12.6	9.8	
		MCWBR	7.4	9.7	12.3	11.4	10.1	9.3	8.7	9.5	10.3	10.5	8.8	8.4	
Clear-Sky Solar Irradiance	taub	0.319	0.350	0.382	0.435	0.409	0.403	0.433	0.432	0.398	0.367	0.346	0.320		
	taud	2.258	2.226	2.240	2.126	2.276	2.310	2.237	2.233	2.298	2.368	2.353	2.320		
	Ebn at Noon	233	251	262	259	271	273	262	257	254	244	222	218		
	Edh at Noon	25	32	37	46	41	40	42	41	35	28	23	21		
All-Sky Solar Radiation	RadAvg	323	580	970	1444	1852	1977	1943	1744	1203	744	335	243		
	RadStd	56	94	116	131	157	134	116	100	162	93	52	53		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		+1.12	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (9 neighbors)		+1.22	N/A	N/A	+1.49	N/A	N/A	-221	-322	+211

Nomenclature: See separate page